

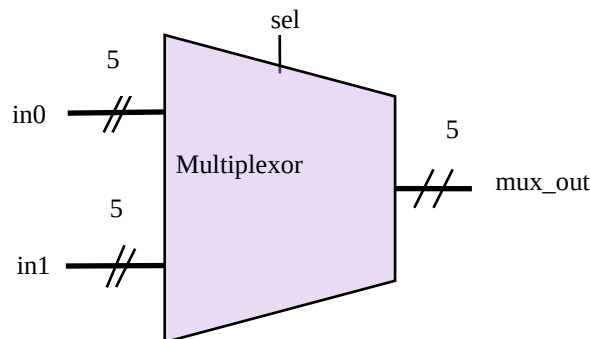
LAB 1 Verilog Introduction

Lab 3-1 Modeling an Address Multiplexor

Objective: To use basic Verilog constructs to describe a simple Multiplexor design.

In this lab, you create a simple multiplexor design using Verilog constructs and verify it using the supplied testbench.

You will create the MUX design as per the specification and verify it using the provided testbench. Please make sure to use the same variable name as the ones shown in block diagrams.



Specifications

- ♦ The address multiplexor selects between the instruction address during the instruction fetch phase and the operand address during the instruction execution phase.
- ♦ MUX width is parameterized with the default value of 5.
- ♦ If `sel` is `1'b0`, input `in0` is passed to the output `mux_out`.
- ♦ If `sel` is `1'b1`, input `in1` is passed to the output `mux_out`.

Designing the MUX

1. Change to the *lab4-mux* directory and examine the following files.

<code>multiplexor_test.v</code>	Multiplexor test
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2. Create the *multiplexor.v* file, and using your favorite editor, describe the multiplexor module named “*multiplexor*”.
3. Parameterize the multiplexor input and output widths and assign a default value of 5.

Verifying the MUX design

1. Using the provided test module, check your multiplexor design using the following command with VCS .

```
vcs multiplexor.v multiplexor_test.v (Batch Mode)
./simv
```

```
[IC@IC Lab 3]$ ./simv
Chronologic VCS simulator copyright 1991-2014
Contains Synopsys proprietary information.
Compiler version I-2014.03; Runtime version I-2014.03; Jul 19 19:53 2023
At time 1 sel=0 in0=10101 in1=00000, mux_out=10101
At time 2 sel=0 in0=01010 in1=00000, mux_out=01010
At time 3 sel=1 in0=00000 in1=10101, mux_out=10101
At time 4 sel=1 in0=00000 in1=01010, mux_out=01010
TEST PASSED
$finish called from file "multiplexor_test.v", line 45.
$finish at simulation time 4
VCS Simulation Report
Time: 4
CPU Time: 31.160 seconds; Data structure size: 0.0Mb
```

or

```
vcs -full64 -debug_access+all multiplexor.v multiplexor_test.v
( GUI Mode)
```

2. You might find it easier to list all the files and simulation options in a text file and pass the file into the simulator using the `-f irun` option.

```
vcs-f filelist.txt -access rwc
```

You should see the following results.

```
At time 1 sel=0 in0=10101 in1=00000, mux_out=10101
At time 2 sel=0 in0=01010 in1=00000, mux_out=01010
At time 3 sel=1 in0=00000 in1=10101, mux_out=10101
At time 4 sel=1 in0=00000 in1=01010, mux_out=01010
TEST PASSED
```

3. Correct your multiplexor design as needed.

